

Smart Recipe Search Engine

An intelligent recipe retrieval and recommendation system based on Information Retrieval techniques.

This project allows users to search recipes using ingredients and retrieve the most relevant recipes using **TF-IDF**, **Cosine Similarity**, and **Popularity Ranking**.

The system provides an interactive Streamlit interface with recipe details, images, calorie filtering, ranking scores, and ingredient-based search.

Features

- Search recipes by ingredients
- TF-IDF based information retrieval
- Cosine Similarity ranking
- Popularity-based recommendation
- Detailed recipe information
- Local recipe image support
- Calorie range filtering
- Rating and review analysis
- Cuisine information
- Interactive Streamlit user interface

Technologies

- Python
- Streamlit
- Pandas
- NumPy
- PyArrow
- Requests
- Information Retrieval Algorithms
- TF-IDF
- Cosine Similarity

Project Structure

```
Recipe_Search_Engine/  
├── app.py  
│   └── Streamlit user interface
```

```
├── build_dataset.py
│   │   Dataset cleaning and preprocessing pipeline
├── download_images.py
│   │   Downloads recipe images locally
├── search_engine.py
│   │   TF-IDF indexing and ranking engine
├── requirements.txt
├── README.md
├── data/
│   ├── recipes_clean.json
│   └── recipes_images.json
├── cache/
│   └── vocabulary.json
├── screenshots/
└── docs/
    └── architecture.md
```

System Architecture

The complete pipeline of the project:

```
Raw Dataset
  |
  v
build_dataset.py
  |
  v
recipes_clean.json
  |
  v
download_images.py
  |
  v
recipes_images.json
  |
  v
search_engine.py
  |
  v
TF-IDF Search Index
  |
```

```
    v
  app.py
    |
    v
Streamlit Application
```

Installation

Clone the repository:

```
git clone repository_url
```

Move into project directory:

```
cd Recipe_Search_Engine
```

Install required packages:

```
pip install -r requirements.txt
```

Running The Project

Option 1: Run Existing Version

If the processed dataset already exists:

```
data/recipes_images.json
```

Simply run:

```
streamlit run app.py
```

The application will automatically create the search index if needed.

Running The Complete Pipeline From Scratch

If you want to rebuild the project completely:

Step 1 - Build Clean Dataset

Run:

```
python build_dataset.py
```

This process:

- Reads the original dataset
- Cleans invalid recipes
- Extracts ingredients
- Processes instructions
- Creates the cleaned dataset

Output:

```
data/recipes_clean.json
```

Step 2 - Download Recipe Images

Run:

```
python download_images.py
```

This step downloads recipe images and creates:

```
data/images/
```

and:

```
data/recipes_images.json
```

Important:

The image folder is not included in the repository because it contains thousands of image files.

To display recipe images, every user must run:

```
python download_images.py
```

before running the application.

Step 3 - Run Application

Start Streamlit:

```
streamlit run app.py
```

Search Algorithm

The ranking system combines three main concepts:

TF-IDF

Recipe documents are converted into numerical vectors using:

- Recipe title
- Ingredients
- Instructions
- Keywords

Cosine Similarity

Measures similarity between the user's query and recipes.

Popularity Score

Uses:

- Recipe rating
- Number of reviews

Final ranking formula:

```
Final Score = 0.9 * Similarity + 0.1 * Popularity
```

Dataset Files

recipes_clean.json

Contains cleaned recipe information:

- Recipe title
- Ingredients
- Instructions

- Ratings
- Calories
- Metadata

recipes_images.json

Contains final processed recipes with local image filenames.

This file is used by the search engine.

Cache Files

The cache folder contains:

vocabulary.json

Contains the generated vocabulary from the TF-IDF model.

This file is lightweight and included in the repository.

index.pkl

Contains the generated TF-IDF search index.

It is automatically created when the application runs for the first time.

The file is intentionally excluded from GitHub because:

- It is a generated cache file
- It can be recreated automatically
- It increases repository size

Screenshots

Home Page

 Home Page

Search Results

Deploy

1. Chicken & Long Grain Rice

55 min

5.0 (55 reviews)

452 kcal

International

Published: June 17, 2002

Similarity: 55.42%

Popularity: 50.14%

Total Score: 54.90%

Total Score: 54.90%



This is one of those dishes that you might find making over and over again. There is so much you can add to this and it still taste great.

Key Ingredients

margarinechickenbreastonionwaterlong grain ricesaltschili powderpepperpaprika

Show All (9 items)

Full Ingredients

- 1 margarine
- 1 chicken breast
- 1 onion

Preparation Steps

Step 1

In a large skillet heat oil and margarine.

Step 2

Add chicken and onion cook and stir until chicken is lightly browned and onions are tender.

2. Cheesy Chicken and Rice Casserole

55 min

4.5 (10 reviews)

319 kcal

International

Published: January 08, 2007

Similarity: 51.23%

Popularity: 35.88%

Total Score: 48.80%

Total Score: 48.80%



Calorie Filtering

Deploy

1. Long Grain and Wild Rice Mix

50 min

5.0 (13 reviews)

196 kcal

International

Published: February 16, 2003

Similarity: 41.60%

Popularity: 32.88%

Total Score: 40.72%

Total Score: 40.72%



A mix that will make a great gift or a pantry favorite. Ever read the additives on a box of this stuff????

Key Ingredients

long grain brown ricewild ricedried parsleydried thymesalt

Show All (5 items)

Full Ingredients

- 1 long grain brown rice
- 1/2 wild rice
- 1 dried parsley

Preparation Steps

Step 1

Mix ingredients and store in an air-tight container.

Step 2

Attach these instructions.

2. Brown Rice & Peppers Pilaf

55 min

5.0 (17 reviews)

182 kcal

International

Published: April 05, 2009

Similarity: 41.24%

Popularity: 36.01%

Total Score: 40.72%



Architecture



Documentation

Technical documentation is available in:

```
docs/architecture.md
```

Notes

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- Original raw datasets are not included because of their large size.
- Recipe images are downloaded locally using `download_images.py`.
- The search index is generated automatically.
- The project can run without images, but recipe images require the image download step.
- `index.pkl` is a generated cache file and should not be uploaded to GitHub.

Author

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